

§3.9.

1. $A = \begin{pmatrix} -3 & 2 \\ -1 & -1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -3-\lambda & 2 \\ -1 & -1-\lambda \end{pmatrix}$$

$$= \lambda^2 + 4\lambda + 5 = 0$$

$$\Rightarrow \lambda_1 = -2+i, \lambda_2 = -2-i$$

ii) $\lambda_1 = -2+i$

$$(A - (-2+i)I)V = 0 \Rightarrow \begin{pmatrix} -1-i & 2 \\ -1 & 1-i \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$(-1-i)V_1 + 2V_2 = 0 \Rightarrow V_1 = (1-i)V_2$$

$$-V_1 + (1-i)V_2 = 0$$

$$\Rightarrow V = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} (1-i)V_2 \\ V_2 \end{pmatrix} = V_2 \begin{pmatrix} 1-i \\ 1 \end{pmatrix} = c \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$$

$$\Rightarrow \tilde{X}(t) = e^{\lambda_1 t} V = e^{(-2+i)t} \begin{pmatrix} 1-i \\ 1 \end{pmatrix} \quad (c=1)$$

$$= e^{-2t} (\cos t + i \sin t) \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$$

$$= e^{-2t} \begin{pmatrix} \cos t + \sin t + i(\sin t - \cos t) \\ \cos t + i \sin t \end{pmatrix}$$

$$\Rightarrow X^1(t) = \operatorname{Re}(\tilde{X}(t)) = e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix}$$

$$X^2(t) = \operatorname{Im}(\tilde{X}(t)) = e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}$$

Thus

$$X(t) = c_1 X^1(t) + c_2 X^2(t)$$

$$= c_1 e^{-2t} \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} \sin t - \cos t \\ \sin t \end{pmatrix}$$

P3. $A = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & -2 \\ 2 & 2 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 0 & 0 \\ 3 & 1-\lambda & -2 \\ 2 & 2 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)((1-\lambda)^2 + 4)$$

$$= (1-\lambda)(\lambda^2 - 2\lambda + 5) = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 1+2i, \lambda_3 = 1-2i.$$

i) $\lambda_1 = 1.$

$$(A - I)V = 0 \Rightarrow \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & -2 \\ 2 & 2 & 0 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 3V_1 - 2V_3 = 0, 2V_1 + 2V_2 = 0$$

$$\Rightarrow V_3 = \frac{3}{2}V_1, V_2 = -V_1$$

$$\Rightarrow V = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} V_1 \\ -V_1 \\ \frac{3}{2}V_1 \end{pmatrix} = V_1 \begin{pmatrix} 1 \\ -1 \\ \frac{3}{2} \end{pmatrix} = c \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$$

$$\Rightarrow X^1(t) = e^t \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$$

ii) $\lambda_2 = 1+2i$

$$(A - (1+2i)I)V = 0 \Rightarrow \begin{pmatrix} -2i & 0 & 0 \\ 3 & -2i & -2 \\ 2 & 2 & -2i \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-2iV_1 = 0 \Rightarrow V_1 = 0$$

$$3V_1 - 2iV_2 - 2V_3 = 0$$

$$2V_1 + 2V_2 - 2iV_3 = 0$$

$$\Rightarrow V = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 0 \\ iV_3 \\ V_3 \end{pmatrix} = V_3 \begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}$$

$$\Rightarrow \tilde{X}(t) = e^{(1+2i)t} \begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}$$

$$= e^t (\cos 2t + i \sin 2t) \begin{pmatrix} 0 \\ i \\ 1 \end{pmatrix}$$

$$\Rightarrow X^2(t) = \operatorname{Re}(\tilde{X}(t)) = e^t \begin{pmatrix} 0 \\ -\sin 2t \\ \cos 2t \end{pmatrix}$$

$$X^3(t) = \operatorname{Im}(\tilde{X}(t)) = e^t \begin{pmatrix} 0 \\ \cos 2t \\ \sin 2t \end{pmatrix}$$

$$\Rightarrow X(t) = c_1 X^1(t) + c_2 X^2(t) + c_3 X^3(t)$$

$$= c_1 e^t \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} + c_2 e^t \begin{pmatrix} 0 \\ -\sin 2t \\ \cos 2t \end{pmatrix}$$

$$+ c_3 e^t \begin{pmatrix} 0 \\ \cos 2t \\ \sin 2t \end{pmatrix}$$

$$= e^t \left(c_1 \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ -\sin 2t \\ \cos 2t \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ \cos 2t \\ \sin 2t \end{pmatrix} \right)$$

P5. $A = \begin{pmatrix} 1 & -1 \\ 5 & -3 \end{pmatrix}$, $x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & -1 \\ 5 & -3-\lambda \end{pmatrix}$$

$$= \lambda^2 + 2\lambda + 2 = 0$$

$$\Rightarrow \lambda_1 = -1+i, \lambda_2 = -1-i$$

i) $\lambda_1 = -1+i$

$$(A - (-1+i)I)v = 0 \Rightarrow \begin{pmatrix} 2-i & -1 \\ 5 & -2-i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$\Rightarrow v_2 = (2-i)v_1$$

$$\Rightarrow v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 \\ (2-i)v_1 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ 2-i \end{pmatrix} = C \begin{pmatrix} 1 \\ 2-i \end{pmatrix}$$

$$\Rightarrow \tilde{x}(t) = e^{(-1+i)t} \begin{pmatrix} 1 \\ 2-i \end{pmatrix}$$

$$= e^{-t} (C_3 t + i \sin t) \begin{pmatrix} 1 \\ 2-i \end{pmatrix}$$

$$= e^{-t} \begin{pmatrix} C_3 t + i \sin t \\ 2C_3 t + \sin t + i(2\sin t - C_3 t) \end{pmatrix}$$

$$\Rightarrow x'(t) = \operatorname{Re}(\tilde{x}(t)) = e^{-t} \begin{pmatrix} C_3 t \\ 2C_3 t + \sin t \end{pmatrix}$$

$$x^2(t) = \operatorname{Im}(\tilde{x}(t)) = e^{-t} \begin{pmatrix} \sin t \\ 2\sin t - C_3 t \end{pmatrix}$$

$$\Rightarrow x(t) = C_1 e^{-t} \begin{pmatrix} C_3 t \\ 2C_3 t + \sin t \end{pmatrix} + C_2 e^{-t} \begin{pmatrix} \sin t \\ 2\sin t - C_3 t \end{pmatrix}$$

From the initial-values,

$$x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + C_2 \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\Rightarrow \begin{cases} C_1 = 1 \\ 2C_1 - C_2 = 2 \end{cases} \Rightarrow \begin{cases} C_1 = 1 \\ C_2 = 0 \end{cases}$$

Thus,

$$x(t) = e^{-t} \begin{pmatrix} C_3 t \\ 2C_3 t + \sin t \end{pmatrix}$$

P7. $A = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix}$, $x(0) = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} -3-\lambda & 0 & 2 \\ 1 & -1-\lambda & 0 \\ -2 & -1 & -\lambda \end{pmatrix}$$

$$= (\lambda+2)(\lambda^2+2\lambda+3)$$

$$\Rightarrow \lambda_1 = -2, \lambda_2 = -1+\sqrt{2}i, \lambda_3 = -1-\sqrt{2}i$$

i) $\lambda_1 = -2$

$$(A+2I)v = 0 \Rightarrow \begin{pmatrix} -1 & 0 & 2 \\ 1 & 1 & 0 \\ -2 & -1 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-v_1 + 2v_3 = 0 \Rightarrow v_3 = \frac{v_1}{2}$$

$$v_1 + v_2 = 0 \Rightarrow v_2 = -v_1$$

$$-2v_1 - v_2 + 2v_3 = 0$$

$$\Rightarrow v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ -v_1 \\ \frac{v_1}{2} \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ -1 \\ \frac{1}{2} \end{pmatrix} = C \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

$$\Rightarrow x'(t) = e^{-2t} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

ii) $\lambda_2 = -1+\sqrt{2}i$

$$(A - (-1+\sqrt{2}i)I)v = 0 \Rightarrow \begin{pmatrix} -2-\sqrt{2}i & 0 & 2 \\ 1 & -\sqrt{2}i & 0 \\ -2 & -1 & 1-\sqrt{2}i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow v_3 = (1+\frac{\sqrt{2}}{2}i)v_1, v_2 = -\frac{\sqrt{2}}{2}i v_1$$

$$\Rightarrow v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ -\frac{\sqrt{2}}{2}i v_1 \\ (1+\frac{\sqrt{2}}{2}i)v_1 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ -\frac{\sqrt{2}}{2}i \\ 1+\frac{\sqrt{2}}{2}i \end{pmatrix}$$

$$= C \begin{pmatrix} \sqrt{2} \\ -i \\ \sqrt{2}+i \end{pmatrix}$$

$$\Rightarrow \tilde{x}(t) = e^{(-1+\sqrt{2}i)t} \begin{pmatrix} \sqrt{2} \\ -i \\ \sqrt{2}+i \end{pmatrix}$$

$$= e^{-t} (C_3 \sqrt{2} t + i \sin \sqrt{2} t) \begin{pmatrix} \sqrt{2} \\ -i \\ \sqrt{2}+i \end{pmatrix}$$

$$= e^{-t} \begin{pmatrix} \sqrt{2} C_3 \sqrt{2} t + i \sqrt{2} \sin \sqrt{2} t \\ \sin \sqrt{2} t - i C_3 \sqrt{2} t \\ \sqrt{2} C_3 \sqrt{2} t - \sin \sqrt{2} t + i(\sqrt{2} \sin \sqrt{2} t + C_3 \sqrt{2} t) \end{pmatrix}$$

$$\Rightarrow x^1(t) = \operatorname{Re}(\tilde{x}(t)) = e^{-t} \begin{pmatrix} \sqrt{2} C_3 \sqrt{2} t \\ \sin \sqrt{2} t \\ \sqrt{2} C_3 \sqrt{2} t - \sin \sqrt{2} t \end{pmatrix}$$

$$x^2(t) = \operatorname{Im}(\tilde{x}(t)) = e^{-t} \begin{pmatrix} \sqrt{2} \sin \sqrt{2} t \\ -C_3 \sqrt{2} t \\ \sqrt{2} \sin \sqrt{2} t + C_3 \sqrt{2} t \end{pmatrix}$$

$$\Rightarrow x(t) = C_1 e^{-2t} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + e^t \left[C_2 \begin{pmatrix} \sqrt{2} C_3 \sqrt{2} t \\ \sin \sqrt{2} t \\ \sqrt{2} C_3 \sqrt{2} t - \sin \sqrt{2} t \end{pmatrix} + C_3 \begin{pmatrix} \sqrt{2} \sin \sqrt{2} t \\ -C_3 \sqrt{2} t \\ \sqrt{2} \sin \sqrt{2} t + C_3 \sqrt{2} t \end{pmatrix} \right]$$

From the initial values,

$$x(0) = \begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix} = C_1 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + C_2 \begin{pmatrix} \sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix} + C_3 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\Rightarrow C_1 = 1, C_2 = -\sqrt{2}, C_3 = -1$$

$$\Rightarrow x(t) = e^{-2t} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + e^t \begin{pmatrix} -2C_3 \sqrt{2} t - \sqrt{2} \sin \sqrt{2} t \\ -\sqrt{2} \sin \sqrt{2} t + C_3 \sqrt{2} t \\ -3C_3 \sqrt{2} t \end{pmatrix}$$